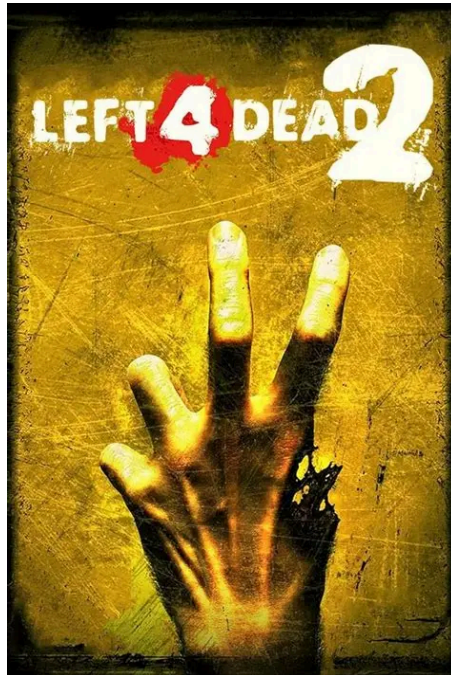


# *Software Requirement Specification*

**By Left4Dead**

**Chosen Irl Challenge: Project D: Course Support & Activity Dashboard**



|                             |            |
|-----------------------------|------------|
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## ***1. Target Users***

- **Lecturer:** Creates a classroom, manages its join code, assigns member roles, creates and updates course activities, posts class announcements, coordinates with TAs and University staff in the staff channel, and reviews the at-risk list. The lecturer is the only role that can promote a member to TA or University staff.
- **Teaching Assistant (TA)/University staff:** A classroom member granted elevated permissions by the lecturer — either a student promoted to TA, or a university staff member added directly to the course in a supporting role. The two are treated as one permission level throughout this document. They help run the course day to day: creating and updating course activities, posting class announcements, coordinating with the lecturer and other staff through the staff-only channel, and monitoring the integrated activity dashboard to track the course's overall progress and rhythm.
- **Student:** Any user who joins a classroom with the join code. Students hold read-only access to class-visible activities and announcements, and may upload a file against an activity that accepts submissions. They cannot post announcements, cannot open the staff channel, and cannot change any activity. All three roles sign in with a Google account, and only accounts in the university's @ku.th domain are admitted.

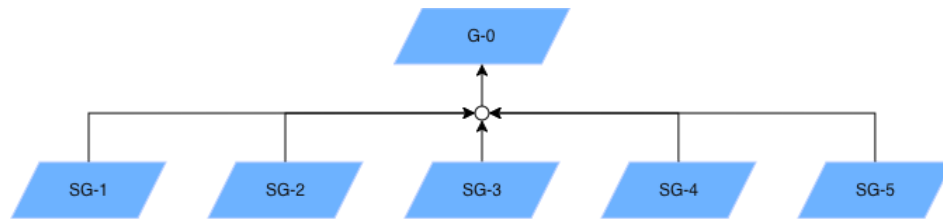
## ***2. The System Goal***

### **G-0: Maintain an accurate and integrated view of course support activities across a semester.**

Course staff currently coordinate assignment deadlines, lab sessions, consultation slots, project milestones, and small internal tasks using a mix of spreadsheets, calendar entries, chat groups, and email threads. No single place shows what is happening when, which work is at risk, or who owns it, and new TAs and University staff have no way to learn the course rhythm.

For this iteration, the system scope is limited to Google sign-in restricted to the @ku.th domain, classroom setup and membership, role assignment, course activity management, an integrated activity dashboard with at-risk detection, class announcements, a staff-only message channel, file submission against an activity deadline, and a read-only student schedule view. Grading, scores, rubrics and feedback on submissions, class-wide chat open to students, person-to-person direct messages, multiple named channels, real-time push delivery, read receipts, and mentions are explicitly out of scope for this iteration and will be addressed in later iterations.

### 3. Goal Refinement via KAOS:



- **Sub-Goal 1 (SG-1)**

**Description:** Maintain: Correct classroom membership and role-based access.

**Rationale:** Access begins at sign-in: authentication is delegated to Google so that no password is stored, and only identities in the university's @ku.th domain are admitted at all. Inside a classroom, a single join code admits everyone as a Student, so the role assignment is the only thing separating read access from write access. Only the lecturer, TAs and University staff of a classroom may change its records, and only the lecturer may promote a member.

- **Sub-Goal 2 (SG-2)**

**Description:** Maintain: Complete and correctly owned course activity records.

**Rationale:** Each activity needs a title, type, due date, status, visibility, and owner, and only authorised staff may create or update one. An activity missing an owner or a due date cannot be tracked or flagged.

- **Sub-Goal 3 (SG-3)**

**Description:** Achieve: Provide one integrated, role-appropriate view of course activities.

**Rationale:** Staff currently reconstruct this view manually from several disconnected tools, and students ask repeated schedule questions because they have no view of their own. Staff see every activity of the classroom; students see only the class-visible ones.

- **Sub-Goal 4 (SG-4)**

**Description:** Avoid: At-risk work and unmet deadlines going unnoticed.

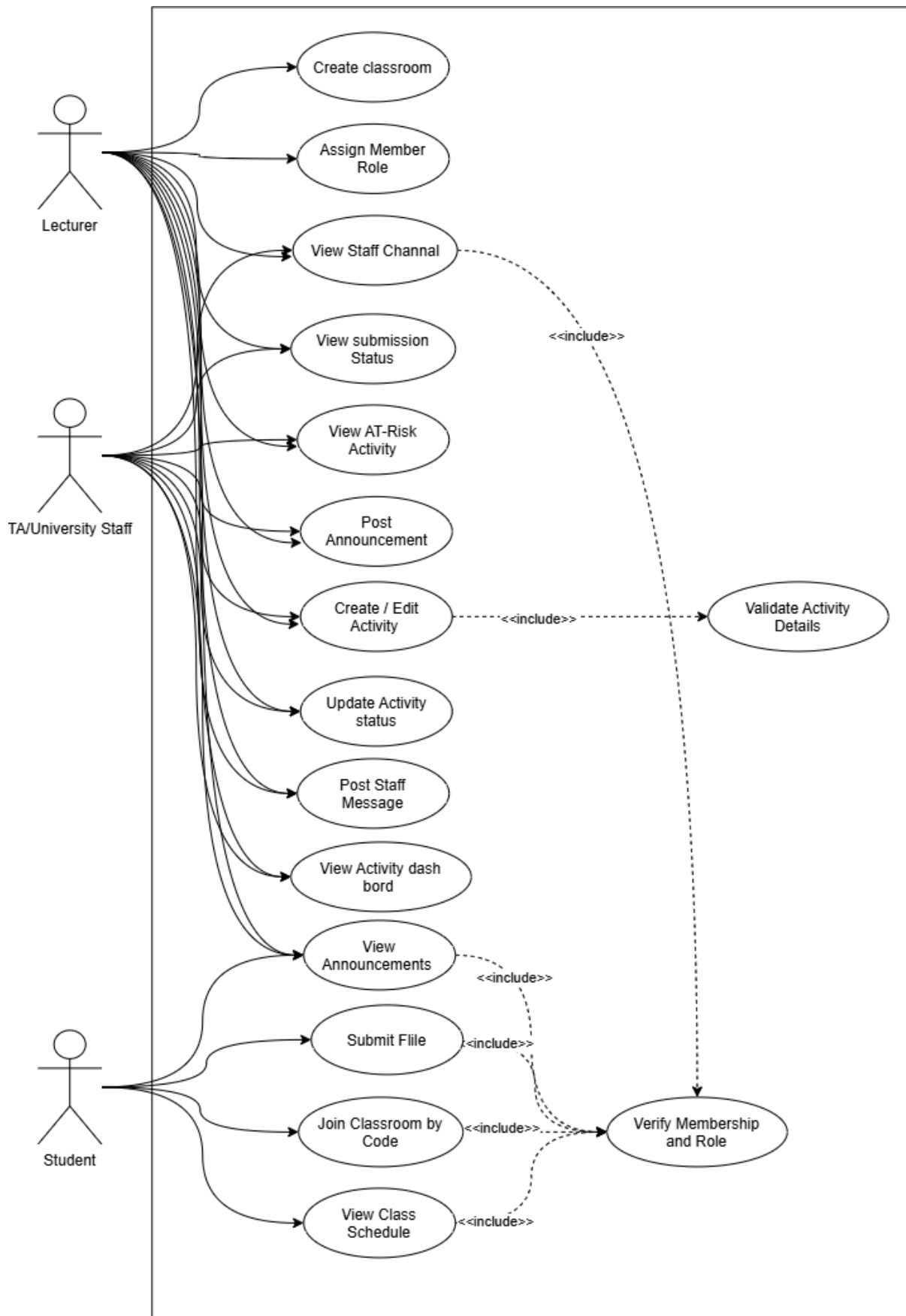
**Rationale:** Overdue, imminent, and unassigned work is the failure the department currently experiences and cannot see. Equally, staff cannot see how many students have submitted before a deadline passes, which is the earliest warning that an assignment is going badly.

- **Sub-Goal 5 (SG-5)**

**Description:** Achieve: Deliver classroom communication to the right audience.

**Rationale:** Notices posted into a chat group scroll away and are missed, so a notice board keeps them readable for the whole semester. Cover swaps, marking questions, and lab problems are a conversation rather than a notice, and must never be visible to students.

#### 4. Use Case Diagram



| <b>ID</b> | <b>Use Case</b>                                                                | <b>Actor</b>                                           |
|-----------|--------------------------------------------------------------------------------|--------------------------------------------------------|
| UC-1      | Create Classroom                                                               | Lecturer                                               |
| UC-2      | Invited user (Lecturer, TA / University staff or Student; admitted as Student) | Student                                                |
| UC-3      | Assign Member Role                                                             | Lecturer                                               |
| UC-4      | Create / Edit Activity                                                         | Lecturer, TA/University staff                          |
| UC-5      | Update Activity Status                                                         | Lecturer, TA/University staff                          |
| UC-6      | View Activity Dashboard                                                        | Lecturer, TA/University staff                          |
| UC-7      | View At-Risk Activities                                                        | Lecturer, TA/University staff                          |
| UC-8      | View Class Schedule                                                            | Student                                                |
| UC-9      | Post Announcement                                                              | Lecturer, TA/University staff                          |
| UC-10     | View Announcements                                                             | Lecturer, TA/University staff, Student                 |
| UC-11     | Post Staff Message                                                             | Lecturer, TA/University staff                          |
| UC-12     | View Staff Channel                                                             | Lecturer, TA/University staff                          |
| UC-13     | Submit File                                                                    | Student                                                |
| UC-14     | View Submission Status                                                         | Lecturer, TA/University staff                          |
| UC-15     | Validate Activity Details                                                      | included by UC-4                                       |
| UC-16     | Verify Classroom Membership and Role                                           | included by UC-2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 |

## 5. User Stories

| User Story ID | Description                                                                                                                                                                                                                                     | Acceptance Criteria                                                                                                                                                |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US-1          | As a lecturer, I want to create a classroom so that a semester's course activities have one place to live.                                                                                                                                      | A classroom is created with a name and semester, and the creating lecturer is recorded as its owner.                                                               |
| US-2          | As a lecturer, I want the system to generate a unique join code so that I can invite TAs, University staff and students without managing accounts manually.                                                                                     | Each classroom has a code that is unique across active classrooms and can be regenerated by the lecturer.                                                          |
| US-3          | As a TA, University staff or student invited to the course, I want to join a classroom using a code so that I can see the activities that concern me. I want to join a classroom using a code so that I can see the activities that concern me. | A valid code adds the user to the classroom as a Student. An invalid code is rejected with a message.                                                              |
| US-4          | As a lecturer, I want to promote a member to TA or University staff so that only the people I trust can change course activities.                                                                                                               | The lecturer can change a member's role between Student, TA and University staff. The member's permissions change on their next request without rejoining.         |
| US-5          | As a lecturer or TA, I want to create an activity with a due date so that responsibility and timing are explicit.                                                                                                                               | Title, due date are required. A saved activity appears in the classroom dashboard.                                                                                 |
| US-6          | As a lecturer, TA or University staff, I want to update a staff activity's status so that the dashboard reflects real progress.                                                                                                                 | Status can be changed between Not Started, In Progress, and Done. The dashboard reflects the change after saving.                                                  |
| US-7          | As a lecturer or TA, I want one integrated view of upcoming activities so that I do not have to check several tools.                                                                                                                            | The dashboard lists all activities of the classroom ordered by due date, showing type, owner, due date, visibility, and status.                                    |
| US-8          | As a lecturer or TA, I want at-risk activities highlighted so that nothing is missed.                                                                                                                                                           | Activities meeting an at-risk rule are visually distinguished and can be listed on their own.                                                                      |
| US-9          | As a student, I want to view the class schedule so that I know when sessions and deadlines fall.                                                                                                                                                | A student sees only activities with Class visibility, and cannot create, edit, or delete any activity.                                                             |
| US-10         | As a lecturer, TA or University staff, I want to post an announcement to the class so that everyone sees the same notice.                                                                                                                       | The announcement is saved with its author and timestamp and becomes visible to every classroom member.                                                             |
| US-11         | As a lecturer, TA, University staff or student in the classroom, I want to read the class announcements in one place so that I do not have to rely on an external chat group.                                                                   | Members see all announcements of the classroom in a single list, ordered newest first. No keyword search or filtering is provided; the list is chronological only. |

| User Story ID | Description                                                                                                                                                                | Acceptance Criteria                                                                                                                                                                                                                                   |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US-12         | As a lecturer, TA or University staff, I want to send a message in the staff channel so that we can settle cover swaps and marking questions without students seeing them. | The message is saved with its author and timestamp. A Student account cannot post to or read the channel.                                                                                                                                             |
| US-13         | As a lecturer, TA or University staff, I want the staff channel to show new messages while I have it open so that a conversation works without reloading the page.         | The channel shows messages oldest to newest and picks up new ones without a manual reload.                                                                                                                                                            |
| US-14         | As a student, I want to upload my file against an assignment so that my work is recorded before the deadline.                                                              | The file is stored with my name, the activity, and the time of upload. Uploading again replaces my previous file.                                                                                                                                     |
| US-15         | As a lecturer, TA or University staff, I want to see how many students have submitted so that I can act before the deadline passes.                                        | The activity shows a submitted count against the class size, and staff can open any student's file.                                                                                                                                                   |
| US-16         | As a lecturer or University staff, I want only Kasetsart University members to be able to log in, so that classroom data stays within the university.                      | Authentication is performed through Google OAuth2. After a successful Google sign-in, the system verifies that the returned email address ends in @ku.th. Accounts with any other domain are rejected with a clear message and no account is created. |



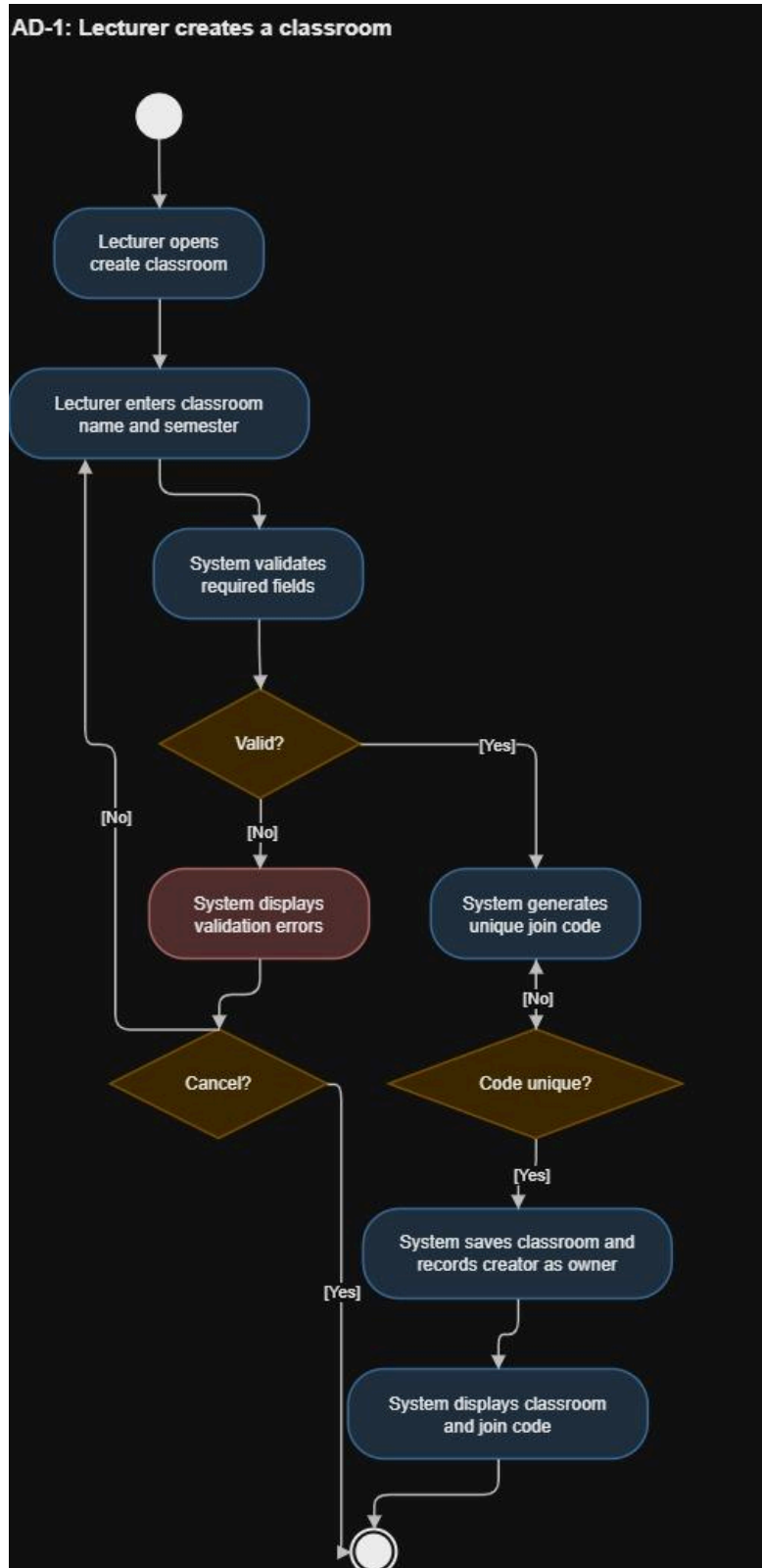
## 6. User Requirement Specification

| ID     | User Requirement                                                                                                                                 |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| URS-1  | A lecturer shall be able to create a classroom and be recorded as its owner.                                                                     |
| URS-2  | The system shall issue a unique join code for each classroom, and the lecturer shall be able to regenerate it.                                   |
| URS-3  | A user shall be able to join a classroom by entering a valid join code, and shall be admitted as a Student.                                      |
| URS-4  | The lecturer who owns a classroom shall be able to change a member's role between Student, TA and University staff.                              |
| URS-5  | Lecturers, TAs and University staff shall be able to create and edit course activities within a classroom they belong to.                        |
| URS-6  | Lecturers, TAs and University staff shall be able to assign an owner and update the status of a course activity.                                 |
| URS-7  | Lecturers and TAs shall be able to view all activities of their classroom in a single view ordered by due date.                                  |
| URS-8  | The system shall identify and highlight activities that are at risk.                                                                             |
| URS-9  | Students shall be able to view activities marked with Class visibility in a classroom they have joined, without modifying them.                  |
| URS-10 | Lecturers, TAs and University staff shall be able to post an announcement to a classroom.                                                        |
| URS-11 | Classroom members shall be able to read the classroom's announcements, most recent first.                                                        |
| URS-12 | Lecturers, TAs and University staff shall be able to exchange messages in a staff channel that students cannot open.                             |
| URS-13 | The staff channel shall show new messages to a member who has it open without requiring a manual reload.                                         |
| URS-14 | Students shall be able to upload a file against an activity that accepts submissions, and shall be able to see and replace their own submission. |
| URS-15 | Lecturers, TAs and University staff shall be able to see how many students have submitted for an activity and to open any submitted file.        |
| URS-16 | Student, Lecturer, TAs and University staff can login by email (Only accept @ku.th)                                                              |

## 7. Activity Diagram

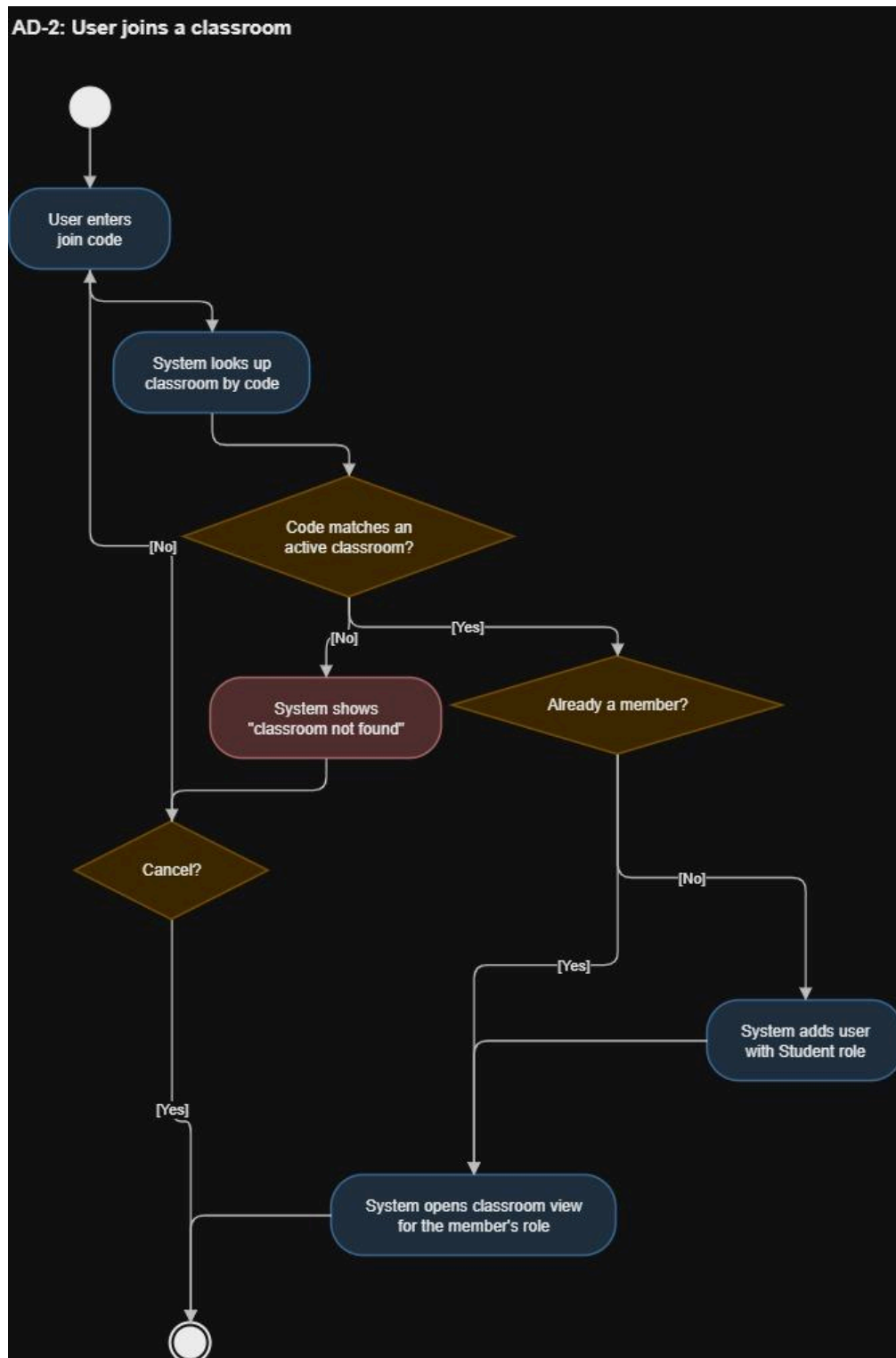
### AD-1: Lecturer creates a classroom

This diagram was used during elicitation to discover the classroom creation and code-generation responsibilities, which were converted into SRS-1, SRS-2, and SRS-3 below.



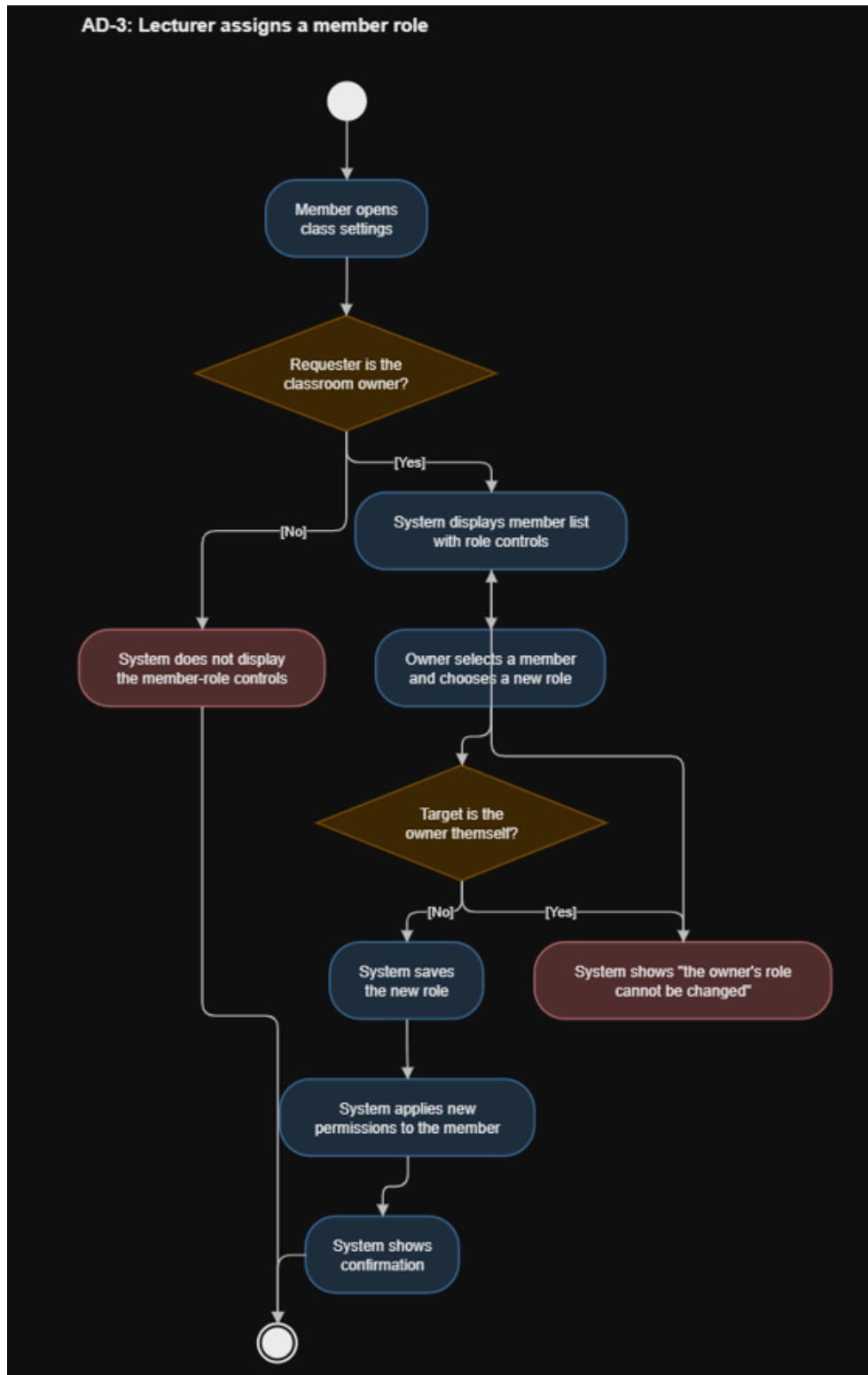
### AD-2: Student, TA or university staff joins a classroom

This diagram captures the membership flow and the default role assignment used to derive SRS-4 and SRS-5.



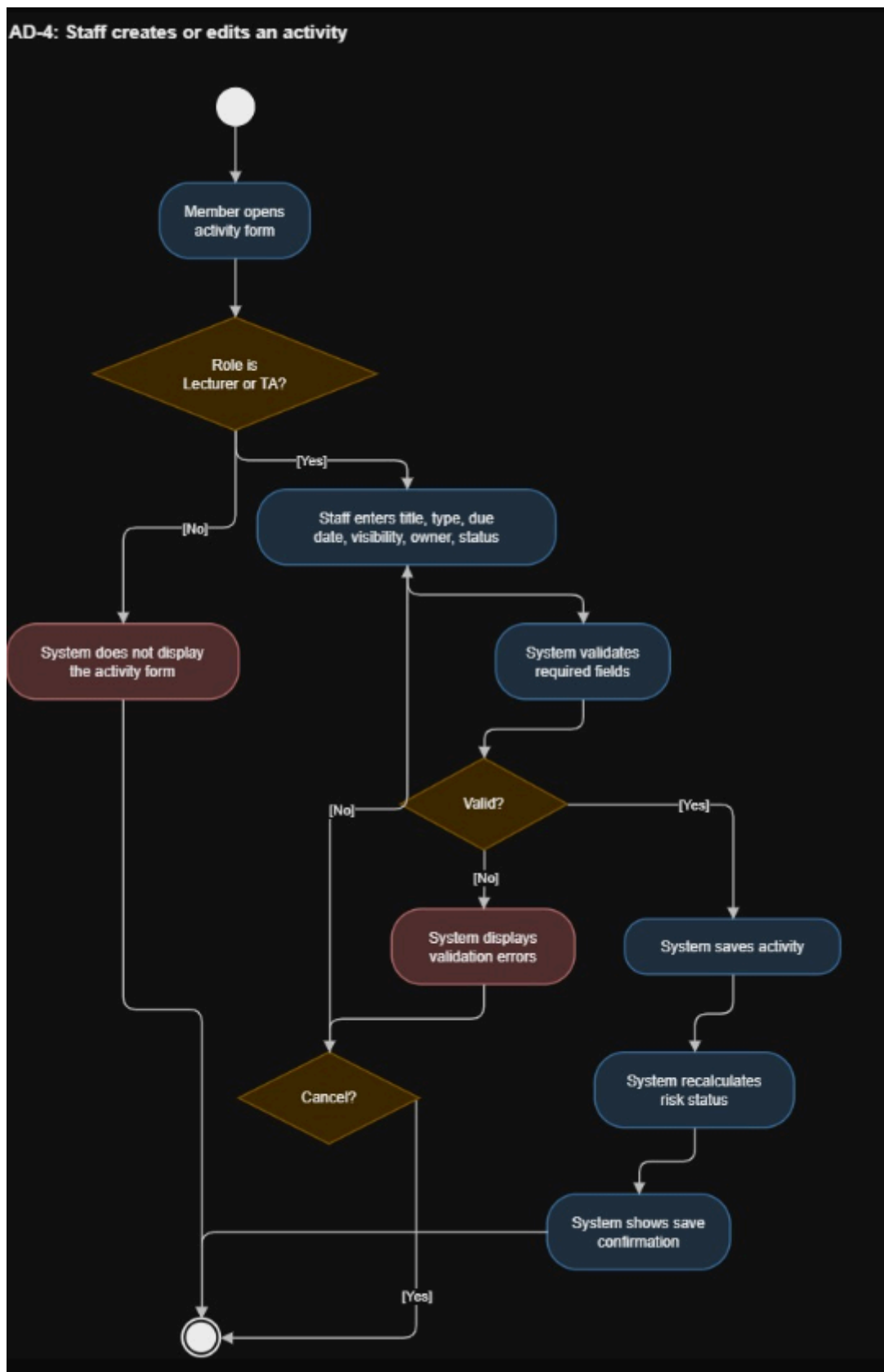
### AD-3: Lecturer assigns a member role

This diagram captures the role-promotion flow used to derive SRS-6 and SRS-7.



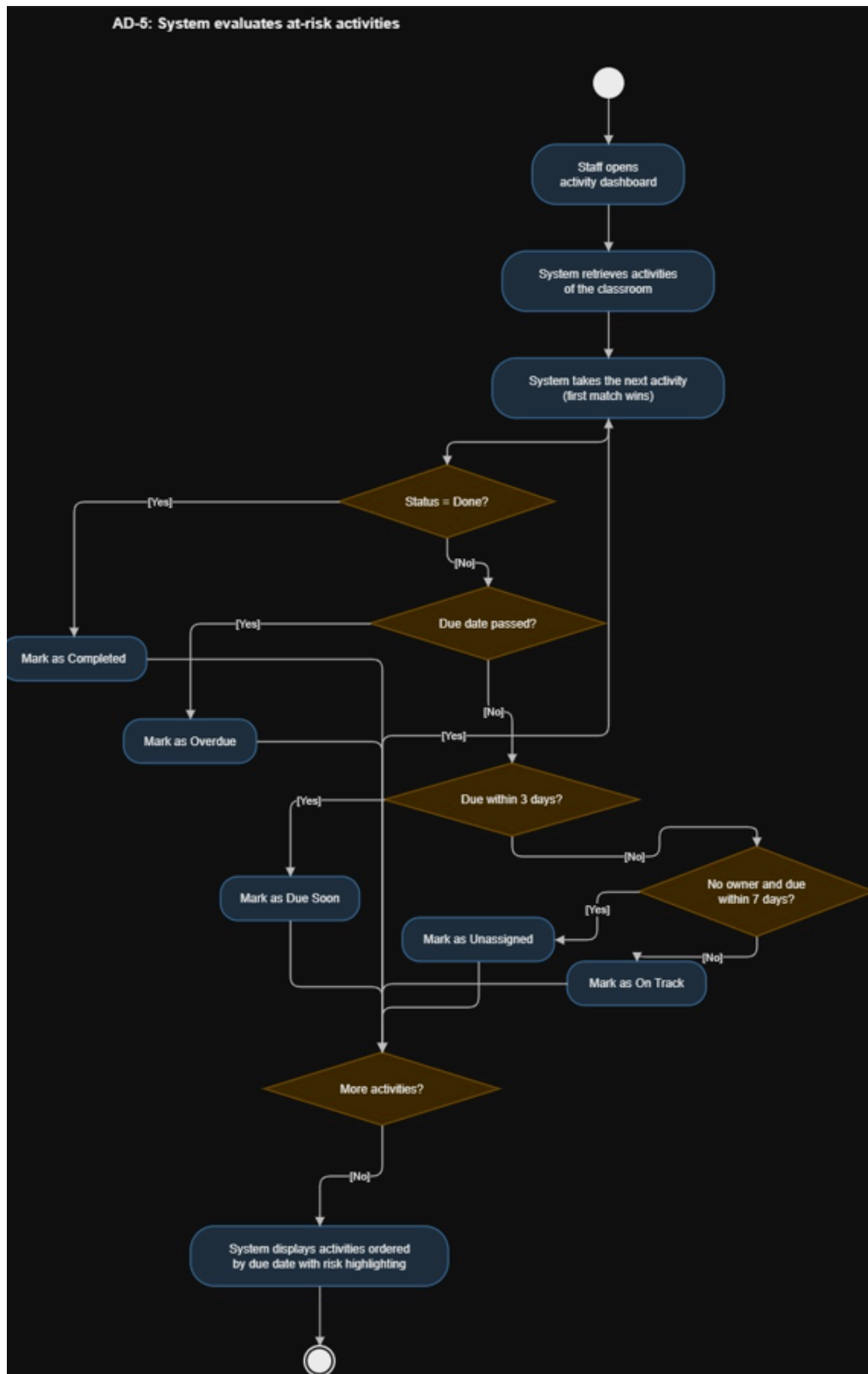
#### AD-4: TA, university staff and lecturer creates or edits an activity

This diagram captures the authorisation check and validation flow used to derive SRS-8 through SRS-11.



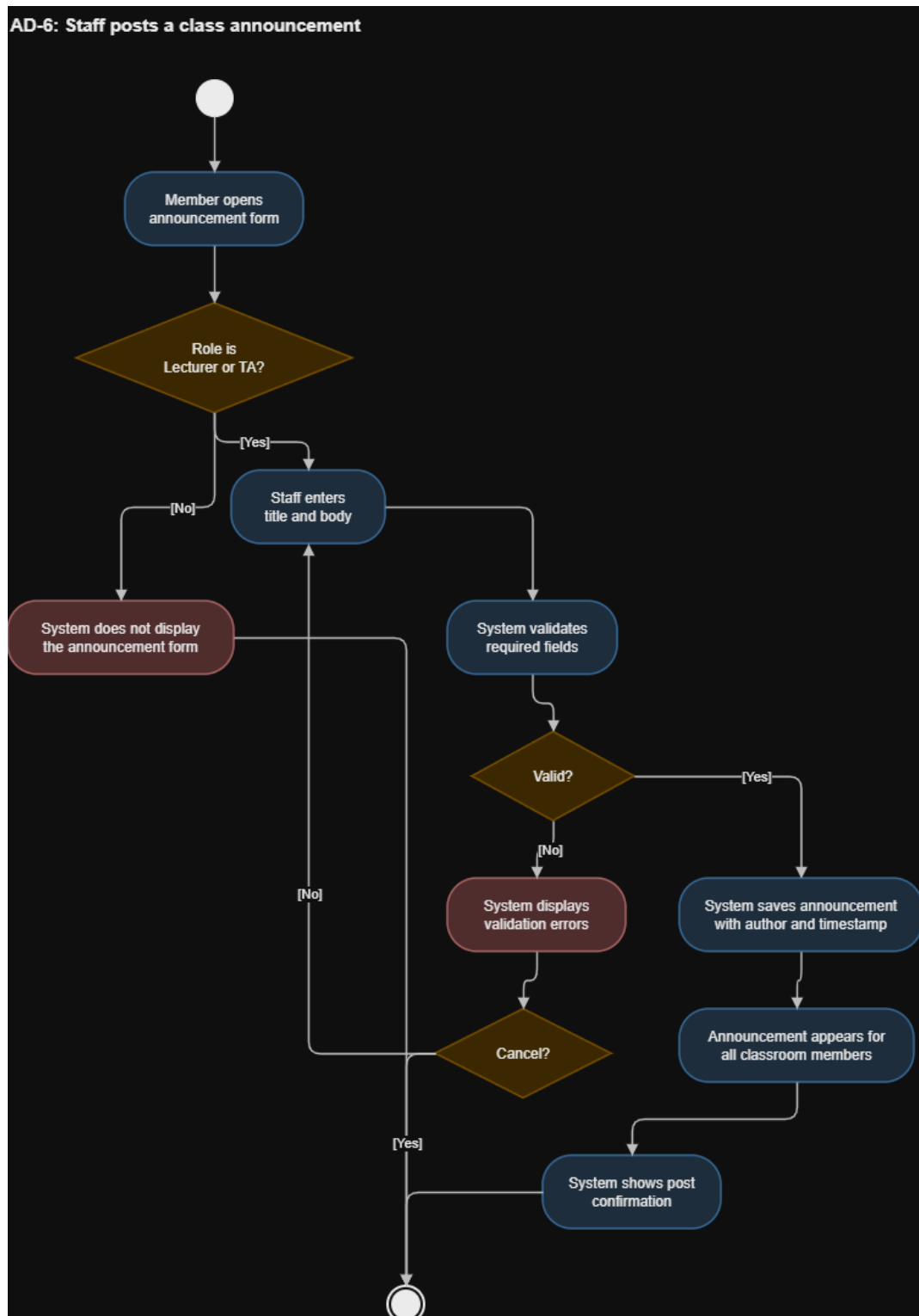
### AD-5: TA and lecturer evaluates at-risk activities

This diagram captures the risk evaluation flow used to derive SRS-12 and SRS-13.



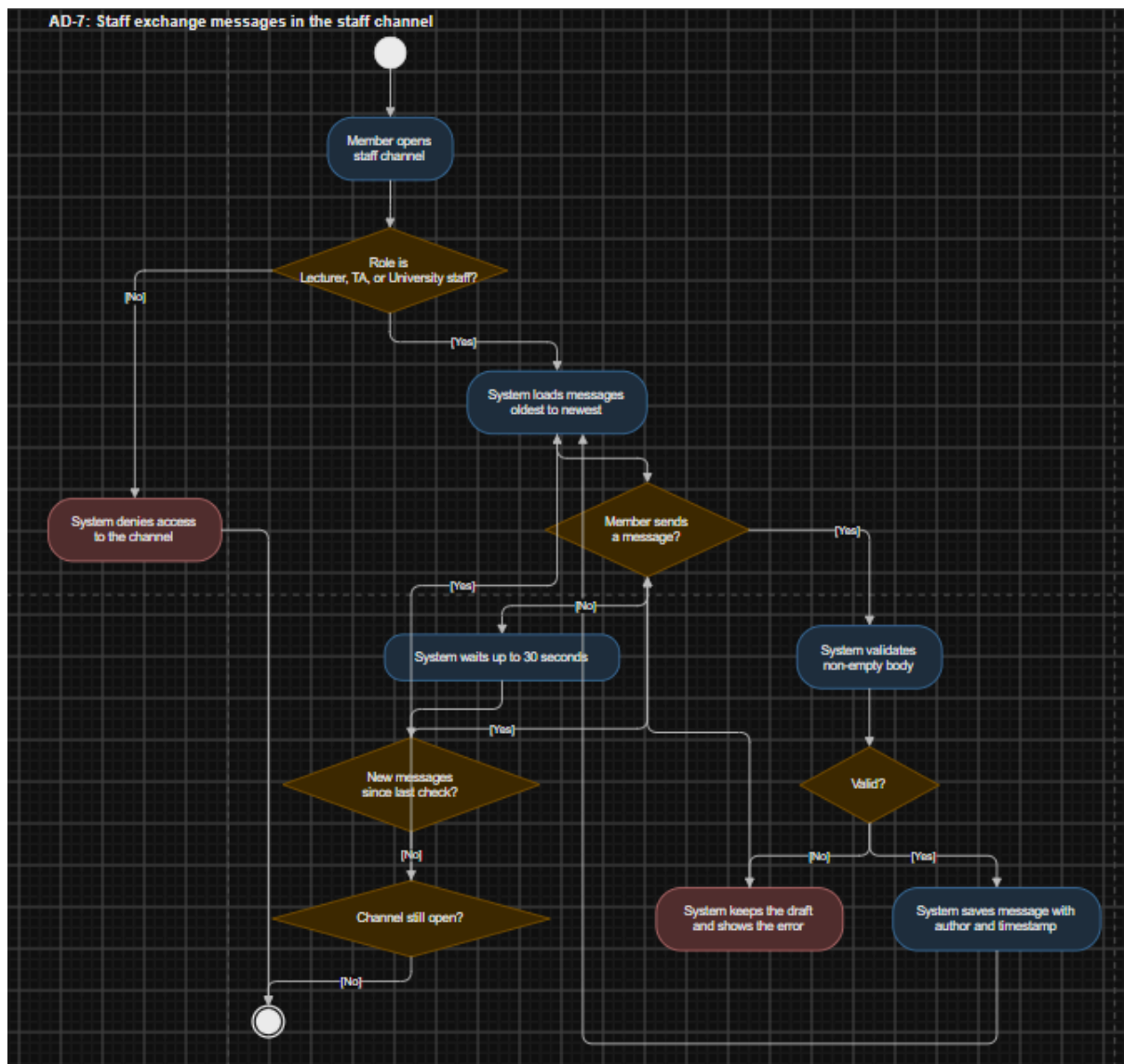
#### AD-6: TA, university staff and lecturer posts a class announcement

This diagram captures the authorisation and validation flow used to derive SRS-15, SRS-16, and SRS-17.



### AD-7: TA, university staff and lecturer exchange messages in the staff channel

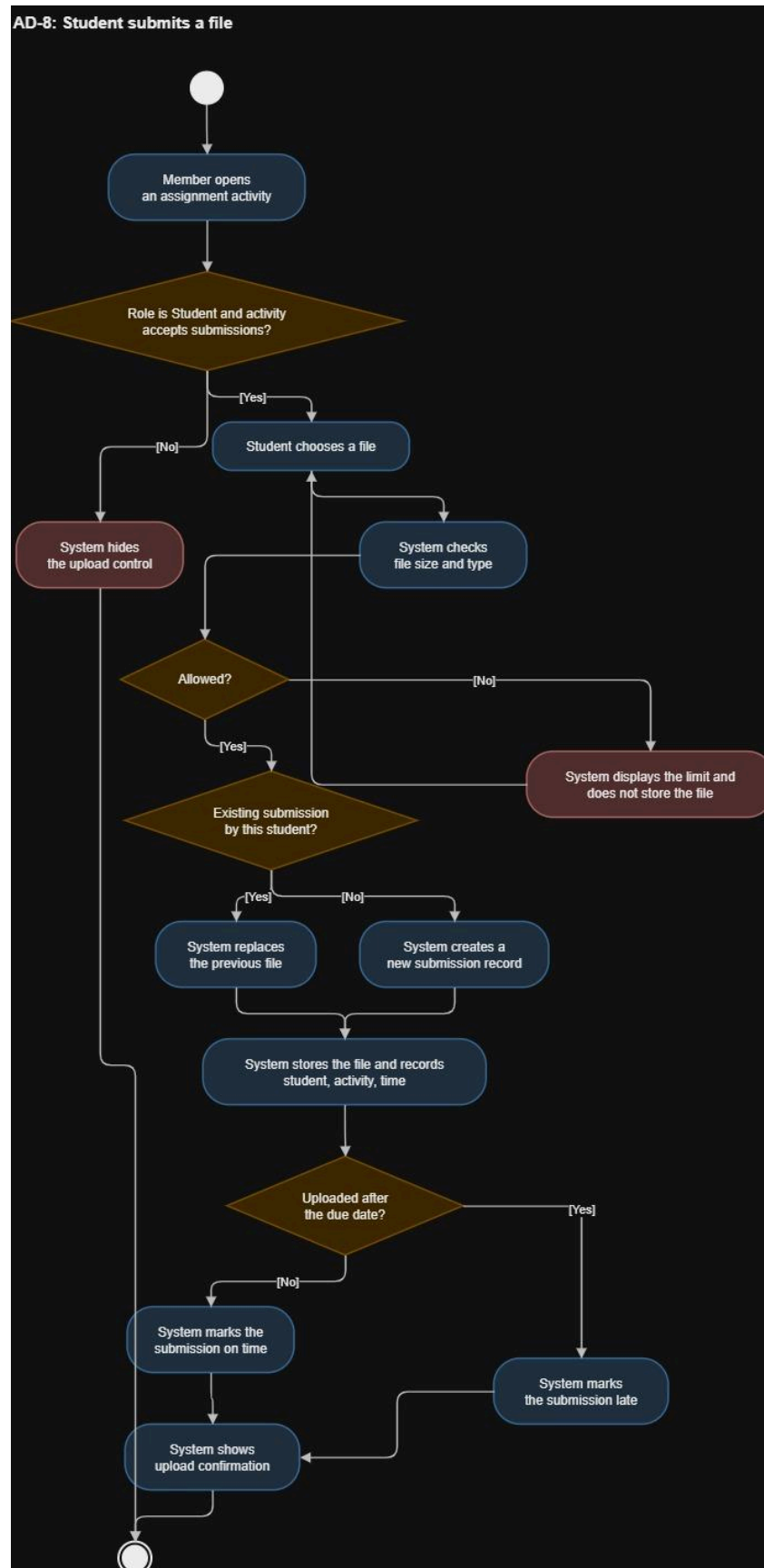
This diagram captures the channel access check and message refresh loop used to derive SRS-18, SRS-19, and SRS-20.





### AD-8: Student submits a file

This diagram captures the deadline check and storage flow used to derive SRS-21 through SRS-25.



## 8. System Requirement Specification

| ID     | System Requirement                                                                                                                                                                                                                                                   |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SRS-1  | The system shall require a classroom name and semester before saving a new classroom.                                                                                                                                                                                |
| SRS-2  | The system shall generate a join code that is unique among all active classrooms.                                                                                                                                                                                    |
| SRS-3  | The system shall record the creating lecturer as the owner of the classroom and shall not allow a non-lecturer account to create a classroom.                                                                                                                        |
| SRS-4  | The system shall add a user to a classroom only when the submitted join code matches an active classroom.                                                                                                                                                            |
| SRS-5  | The system shall assign the Student role to every user admitted through a join code.                                                                                                                                                                                 |
| SRS-6  | The system shall allow only the classroom owner to change a member's role between Student, TA and University staff, and shall not allow the owner's own role to be changed.                                                                                          |
| SRS-7  | The system shall evaluate every request against the member's current role, so that a role change takes effect without the member rejoining the classroom.                                                                                                            |
| SRS-8  | The system shall allow only members with the Lecturer, TA or University staff role to create, edit, or delete activities in that classroom.                                                                                                                          |
| SRS-9  | The system shall require an activity title, activity type, due date, and visibility of Class or Staff before saving a new or edited activity.                                                                                                                        |
| SRS-10 | The system shall display a validation message and shall not save the activity when required data is missing or invalid.                                                                                                                                              |
| SRS-11 | The system shall store each activity with a status of Not Started, In Progress, or Done, and an optional owner.                                                                                                                                                      |
| SRS-12 | The system shall classify an activity as Overdue when its due date has passed and its status is not Done; as Due Soon when its due date falls within 3 days and its status is not Done; and as Unassigned when it has no owner and its due date falls within 7 days. |
| SRS-13 | The system shall display activities of a classroom ordered by due date with their risk classification, and shall provide a filtered list containing only activities classified as Overdue, Due Soon, or Unassigned.                                                  |
| SRS-14 | The system shall display to a Student only activities with Class visibility, and shall reject any create, edit, or delete request from a Student account.                                                                                                            |
| SRS-15 | The system shall allow only members with the Lecturer, TA or University staff role to post an announcement in that classroom.                                                                                                                                        |
| SRS-16 | The system shall require an announcement title and body before posting, and shall record the author and the time of posting.                                                                                                                                         |
| SRS-17 | The system shall display a classroom's announcements to every member of that classroom, ordered most recent first.                                                                                                                                                   |
| SRS-18 | The system shall provide each classroom with one staff channel, and shall reject any request to read or post in it from a member who does not hold the Lecturer/TA/University staff role.                                                                            |
| SRS-19 | The system shall store each staff message with its author, body, and time of sending, and shall reject a message with an empty body.                                                                                                                                 |

| ID     | System Requirement                                                                                                                                                                                                                                               |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SRS-20 | The system shall display staff messages oldest to newest and shall deliver messages sent by other members to an open channel within 30 seconds without a manual page reload.                                                                                     |
| SRS-21 | The system shall accept a file upload from a Student only for an activity marked as accepting submissions in a classroom the student has joined.                                                                                                                 |
| SRS-22 | The system shall reject a file that exceeds the configured maximum size or is not of an allowed type, and shall display the applicable limit.                                                                                                                    |
| SRS-23 | The system shall store each submission with the submitting student, the activity, the original file name, and the time of upload, and shall replace a student's previous submission for the same activity when they upload again.                                |
| SRS-24 | The system shall mark a submission as late when its upload time is after the activity's due date.                                                                                                                                                                |
| SRS-25 | The system shall allow a Student to open only their own submission, shall allow Lecturers and TAs/University staff to open any submission in their classroom, and shall display the number of students who have submitted against the number of Student members. |
| SRS-26 | The system shall delegate user authentication to Google OAuth2 and shall not store user passwords.                                                                                                                                                               |
| SRS-27 | The system shall reject any authenticated identity whose email domain is not @ku.th, returning an access-denied response.                                                                                                                                        |

| Sub-Goal | Use Case                                                                                                                | User Story                 | URS                            | Activity Diagram | SRS                                                    |
|----------|-------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------------|------------------|--------------------------------------------------------|
| SG-1     | UC-1 Create Classroom, UC-2 Join Classroom by Code, UC-3 Assign Member Role, UC-16 Verify Classroom Membership and Role | US-1, US-2, US-3, US-4     | URS-1, URS-2, URS-3, URS-4     | AD-1, AD-2, AD-3 | SRS-1, SRS-2, SRS-3, SRS-4, SRS-5, SRS-6, SRS-7        |
| SG-2     | UC-4 Create / Edit Activity, UC-5 Update Activity Status, UC-15 Validate Activity Details                               | US-5, US-6                 | URS-5, URS-6                   | AD-4             | SRS-8, SRS-9, SRS-10, SRS-11                           |
| SG-3     | UC-6 View Activity Dashboard, UC-8 View Class Schedule                                                                  | US-7, US-9                 | URS-7, URS-9                   | AD-4, AD-5       | SRS-13, SRS-14                                         |
| SG-4     | UC-7 View At-Risk Activities, UC-13 Submit File, UC-14 View Submission Status                                           | US-8, US-14, US-15         | URS-8, URS-14, URS-15          | AD-5, AD-8       | SRS-12, SRS-13, SRS-21, SRS-22, SRS-23, SRS-24, SRS-25 |
| SG-5     | UC-9 Post Announcement, UC-10 View Announcements, UC-11 Post Staff Message, UC-12 View Staff Channel                    | US-10, US-11, US-12, US-13 | URS-10, URS-11, URS-12, URS-13 | AD-6, AD-7       | SRS-15, SRS-16, SRS-17, SRS-18, SRS-19, SRS-20         |

## 9. Software Architecture

Based on the SRS for the Course Support & Activity Dashboard (Left4Dead2), the recommended architecture is a **Modular Monolithic Architecture** using the **Model-View-Controller (MVC)** pattern. This is a single deployable application internally organized into clearly separated modules and layers, avoiding the operational overhead of microservices while preventing the code tangling of a flat monolith.

### Rationale

The project supports collaborative course management with distinct user roles (Lecturer, TA/university staff, and Student) and real-time communication needs for staff. However, as a student group project handling a single university course's activity tracking, announcements, and staff coordination, a full microservices architecture would introduce unnecessary complexity (such as service discovery and inter-service network management).

An MVC modular approach allows a clean separation of concerns:

- **Presentation (View):** Delivers role-appropriate user interfaces for Students, TAs, University stuff, and Lecturers.
- **Controller:** Manages request routing, permission checks, and membership validation (Verify Classroom Membership and Role).
- **Model / Business Logic:** Encapsulates business rules such as risk-status evaluation (Overdue, Due Soon, Unassigned), classroom join-code generation, and role-promotion constraints.

| Component                                | Layer/Type          | Description                                                                                                                                                                             |
|------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Web/Mobile Client</b>                 | Presentation (View) | Renders the class schedule, dashboards, announcement boards, and file upload forms tailored to the active user's role (Student, TA, or Lecturer).                                       |
| <b>Classroom and Activity Controller</b> | Controller          | Handles HTTP requests for classroom setup, role management, activity creation, risk-status filtering, and message posting; performs authorization checks before invoking business logic |
| <b>Course Model / Business Logic</b>     | Model               | Encapsulates core rules including unique join-code validation, role-permission rules, and automatic risk classification algorithms (Overdue, Due Soon, Unassigned).                     |
| <b>Course Database</b>                   | Persistence         | Stores structured relational data for users, memberships, roles, activities, announcements, staff messages, and file submission metadata.                                               |
| <b>Authentication &amp; Role Module</b>  | Cross-cutting       | Verifies classroom membership and checks whether a user holds Student, TA, or Lecturer privileges for every secured request.                                                            |